

Electromagnetism Demystified

emergent properties of the luminiferous aether

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Dr Michael Heffron developed these slides for presentation during Aether Round Table 144 discussion (available on the Space Audits channel of YouTube, https://www.youtube.com/live/2b0gJorxZyQ?si=EUCLW5_y9bpZSDik), and hereby gives permission to freely distribute these presentation slides.

I'd like to thank Alan and Eric for their suggestions for how to improve this presentation, and all of the other Aether Cosmology lads & ladettes for their encouragement and support.

Please

Interrupt me if my words don't make sense to you!

The purpose of this presentation is to communicate some recent discoveries.

I'm doing my best to present this material in an order that will be easy to follow, but what is obvious to me isn't always obvious to everyone else.

I've tried to choose my words carefully, but, if my words don't make sense to you, please interrupt as much as necessary to ask for better explanations until everything makes sense. Everyone will benefit from a discussion of whatever anyone finds confusing, so please participate!

Mystery box

What's inside?

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Half a century ago, a laboratory assignment for my freshman chemistry course gave us mystery boxes.

We had to figure out what was inside the box based on things we could observe from the outside.

For example, we could tilt the box to deduce whether the contents rolled, slid, tumbled, etc.

When the contents hit the side of the box, we could infer whether the collision was elastic or inelastic.

That lab was all about making inferences and deductions from whatever clues we could gather.

Not just chemistry, but science in general, is all about discovering what you know, despite what you don't.

Experiments are important

Imagination

Mathematical derivations

Physical demonstrations or measurements

Best when duplicated and peer reviewed

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Experiments can take on many forms.

Click 1: In his mind, Albert Einstein rode on a beam of light.

Click 2: On paper, Oliver Heaviside reorganized and unified the efforts of Coulomb, Kelvin, and Maxwell.

Click 3: Using a glass of water, Tayler demonstrated that a refracted knife doesn't block a real laser beam.

Click 4: Experiments are most valuable when they are duplicated and critiqued.

Whatever method you choose, it's important to do your own experiments and reach your own conclusions.

What is the luminiferous aether?

A substance similar to neutrinos

No electromagnetic attraction/repulsion

Completely elastic collisions

Superfluid ($v=\eta=0$), isothermal ($\Delta T=0$)

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The luminiferous aether is unmistakably the primary substance of the universe, but we are still discovering what the aether is exactly. As with everything in science, we have to focus on what we can know despite what we don't know.

Click 1: My decades of studying the luminiferous aether have taught me that the most accurate analogy is to view the aether as a universe full of neutrinos. Like neutrinos, imagine the substance of the aether is a million times smaller than electrons. That would cause the aether to behave similar to a gas.

Click 2: Neutrinos are electromagnetically neutral, so you should infer there is no mutual attraction or repulsion.

Click 3: In the absence of compelling evidence to the contrary, imagine that collisions among the aether substance is completely elastic (meaning it just bounces off itself and keeps going).

Click 4: Because the collisions are elastic and there is no attraction or repulsion, you should deduce that the aether is a zero viscosity superfluid. In other words, there is no kinematic or dynamic viscosity (because $\eta = \rho v$). Superfluids are known to have no temperature gradients, because they conduct heat away instantaneously.

Perspectives



↑ dimension ×	× velocity →			
	mass (kg)	momentum (kg·m/s)	energy (kg·m ² /s ²)	
	linear density (kg/m)	mass flow (kg/s)	force (kg·m/s ²)	
	area density (kg/m ²)	viscosity (kg/m·s)	surface tension (kg/s ²)	
	density (kg/m ³)	mass flux (kg/m ² s)	pressure (kg/m·s ²)	

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Before moving on, I want to remind everyone that the water in Newton's bucket simultaneously has all of the attributes depicted by this table.

Table 2-1 from my book, *The Luminiferous Aether: primary substance of the universe*.

The water in the bucket has mass, momentum, energy, viscosity, surface tension, density, mass flux, pressure, etc. All of those are simultaneously valid attributes of the single volume of water in the bucket. Spinning the bucket doesn't change any of those attributes.

That all of those attributes are valid simultaneously is critically important to remember when you are discussing various topics with people who tend to derail every conversation by continuously changing the frame of reference. Failing to realize that all frames of reference simultaneously exist is one way many people confuse themselves and other people.

CLICK: Power was not originally in this table, but I added it off to the right because the next slide will discuss the discovery that electromagnetic and physical measurements converge at the frame of reference we call power.

Watt, Ohm, & Joule

1765: Watt improved how steam engines work

1827: Ohm discovered that $V = I \times R$

1841: Joule discovered that $P = I \times V = I^2 \times R$ (Watt's Law)

1850s: electrical work = mechanical (physical) work

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What I'm about to tell you are some related historical events, which are essential for understanding the luminiferous aether.

Click 1: In 1765, James Watt invented a device called the "separate condenser". It conserved steam power and reduced fuel consumption by approximately 75% for steam engines. It was Watt who gave us our basic understanding of the physics concept of work.

Click 2: In 1827, Georg Simon Ohm discovered that voltage is equal to current intensity multiplied by resistance. We're going to revisit this again soon.

Gemini AI told me the letter I comes from the French phrase intensité du courant, which translates to "current intensity". Gemini further explained that the letter C was already used for capacitance, and it was thought that using the letter A could cause confusion between the concept of current and the ampere unit of measurement. So, for anyone who was curious, that is how the letter I came to represent current.

Click 3: In 1841, James Prescott Joule discovered that the heat generated in a wire is proportional to the current intensity multiplied by the voltage ($P = IV = I^2R$). It was Joule who realized that electrical work was related to physical work, but the law was named to honor Watt's discoveries related to work.

Click 4: Prior to the 1850s, the different units for mechanical and electromagnetic measurements created considerable confusion. Various engineers and scientists contributed to the realization that mechanical horsepower and electrical power represent the same physical reality. That is what caused their convergence at power in the prior table.

Meter-kilogram-second-ampere (MKSA)

1901: Giovanni Giorgi proposed MKS

1948: Système International (SI) added the ampere (MKSA)

Coulombs were defined as ampere·seconds ($C=A\cdot s$)

2019: redefined its base units

Ampere tied to elementary charge, not magnetic force

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Click 1: In 1901, Giovanni Giorgi proposed the meter-kilogram-second (MKS) measurement units to replace the slightly older centimeter-gram-second (CGS) system. At that time, Giorgi proposed adding electromagnetic units, but the world wasn't ready yet for consensus on what those new-fangled electromagnetic measurement units should be.

Click 2: It took the Système International (SI) until 1948 to agree on the ampere as the proper electromagnetic base unit to upgrade its MKS system into MKSA. In turn, that finally made it possible for the world to standardize electromagnetic measurement units. The world is still in a state of such confusion to this day because those developments were so recent.

Click 3: In the past, I always expressed electromagnetic measurements in terms of coulombs, which are ampere·seconds. That was a mistake because I should have been converting electromagnetic units into physical units by eliminating only the amperes rather than ampere·seconds (coulombs).

Click 4: When SI redefined their base units in 2019, it became especially important to separate amperes from ampere·seconds.

Click 5: Because amperes are now tied to the elementary charge (which is ampere·seconds) rather than to magnetic force (which is force per ampere·squared), that change introduced a nearly imperceptible timing error into conversions between electromagnetic units (MKSA) and physical units (MKS).

Luminiferous aether

Kepler's Third Law reduces to constant $v^2 r$ for satellites

That scales down to the subatomic level

At the “luminiferous radius”, velocity is the speed of light

$$c = \sqrt{\frac{P}{\rho}}$$

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Now we will begin discussing how the luminiferous aether ties all of this together.

Click 1: Kepler's Third Law reduces to a constant velocity squared times radius of orbit for each satellite of any given body.

Click 2: That scales down to the subatomic level.

Click 3: At the “luminiferous radius” of a body, its satellites reach the “speed” of light. Like sound, light is a wave that propagates, so it is more correct to call that a rate of propagation rather than a speed.

Click 4: In his book *Philosophiæ Naturalis Principia Mathematica*, Sir Isaac Newton explained that the **elasticity** of an isothermal fluid ($\Delta T=0$) is equal to its pressure. Bulk modulus is Newton's more general terminology for elasticity. Because the luminiferous aether is an isothermal superfluid, the rate of propagation of its waves is equal to the square root of the aether's (pressure divided by its density). This is Newton's equation for the rate of sound propagation, which is only accurate for isothermal fluids. It was slightly inaccurate for sound waves in air, because air isn't isothermal. Because the luminiferous aether is an isothermal fluid, this equation very accurately determines the rate of light propagation in the aether.

BTW, this is the null hypothesis that must be refuted for Newton's theory of gravity and Einstein's theory of relativity to be correct. Because this null hypothesis (that Kepler's Third Law scales down to the subatomic level) is demonstrably true without gravitational or relativistic corrections, those hypotheses are both incorrect.

Python script (GitHub)

github.com/DrMikeHeffron/LuminiferousAether/blob/main/aether.py

MKS are physical measurement units

distance, mass, time

MSKA adds electromagnetic measurement units

volts, amperes, ohms, henries, farads, teslas, etc.

BUT becomes physical when the A is missing

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The math used during this presentation is intentionally simplified.

Click 1: For anyone who wants to take a really deep dive into the math, you can find a Python script in my GitHub repository that will let you easily examine and manipulate the electromagnetic and physical properties of the luminiferous aether.

Click 2: Meters, kilograms, and seconds are physical measurement units

Click 3: of distance, mass, and time.

Click 4: Adding amperes to physical units converts them into electromagnetic measurements

Click 5: of volts, amperes, ohms, henries, farads, teslas, etc.

Click 6: All that is necessary to convert electromagnetic measurements (MKSA) into physical measurements (MKS) is to eliminate the amperes. Similarly, you only need to add the amperes back to convert physical measurements (MKS) into electromagnetic measurements (MKSA).

My Python script does that, for anyone who wants to play with the software.

Surface charge density of the aether (MKSA↔MKS)

$$\sigma^2 \equiv \rho / \mu_0$$

⇓

$$\rho \equiv \mu_0 \sigma^2$$

⇓

$$\mu_0 \equiv \rho / \sigma^2$$

σ is surface charge density
 ρ is aether density
 μ_0 is magnetic permeability

$$\rho = \frac{m_p}{\frac{4\pi}{3} \left(\frac{E_h a_0}{m_e c^2} \right)^3}$$

m_p is proton mass
 E_h is Hartree energy
 a_0 is Bohr radius
 m_e is electron mass
 c is propagation rate of light

‘Circular’ definition required due to 2019 ampere redefinition

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Surface charge density of the luminiferous aether is what permits electromagnetic units (MKSA) to be converted to physical units (MKS), and vice versa.

Electromagnetic units are blue on this slide to make them easy to identify.

Click 1: Surface charge density squared is the density of the aether divided by its magnetic permeability. That is an exact identity, which is necessary specifically because SI redefined the ampere in 2019.

Click 2-3: Because of the identity, the density of the aether is its magnetic permeability multiplied by its squared surface charge density.

Click 4-5: To complete this sequence, magnetic permeability is the density of the aether divided by its squared surface charge density.

Click 6: There are numerous other (slightly less accurate) ways to define all of these quantities. This is merely the most accurate method due to SI's 2019 redefinition of the ampere.

Surface charge density of the aether (MKSA)

$$\rho \quad / \quad \mu_0 \quad = \quad \sigma^2$$

$$\text{m}^{-3}\text{k}^1\text{s}^0\text{A}^0 \quad / \quad \text{m}^1\text{k}^1\text{s}^{-2}\text{A}^{-2} \quad = \quad \text{m}^{-4}\text{k}^0\text{s}^2\text{A}^2$$

$$\rho \quad \times \quad \mu_0^{-1} \quad = \quad \sigma^2$$

$$\text{m}^{-3}\text{k}^1\text{s}^0\text{A}^0 \quad \times \quad \text{m}^{-1}\text{k}^{-1}\text{s}^2\text{A}^2 \quad = \quad \text{m}^{-4}\text{k}^0\text{s}^2\text{A}^2$$

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Click 1-2: I added an electromagnetic units analysis here to clarify what MKSA units these terms represent, and to make it obvious how the units transition during the division.

Click 3-5: For those of you who may not know or remember the rules for exponents: Division by exponents takes a reciprocal by simply flipping the polarity of each exponent. Then you can multiply that reciprocal by just adding up the exponents for each measurement unit. That is a trivially easy method to avoid cancellations and/or messy rearrangement of the reciprocal.

This type of units analysis will come up a few more times during this presentation, so I just want to make sure everyone understands that the Negative Exponent Rule allows you to divide by reversing the polarity of the exponents and then multiply. That's what I did to go from division by μ_0 on top to multiplication by μ_0^{-1} on the bottom. Any variable that doesn't have an exponent, like ρ and μ_0 , has an assumed exponent of 1.

It's also important to understand that multiplying only requires summing the exponents for each measurement unit. That's what I did by adding -3 of ρ to the -1 of μ_0^{-1} to find that the meters of σ^2 have an exponent of -4. Similarly, 1 plus -1 is 0 for the kilograms, 0 plus 2 is 2 for the seconds, and 0 plus 2 is 2 for the amperes. Are there any questions about any of this before we move on?

Surface charge density of the aether (MKSA $\leftrightarrow$ MKS)

$$\rho \quad / \quad \mu_0 \quad = \quad \sigma^2$$

$$\text{m}^{-3}\text{k}^1\text{s}^0\text{A}^0 \quad / \quad \text{m}^1\text{k}^1\text{s}^{-2}\text{A}^{-2} = \text{m}^{-4}\text{k}^0\text{s}^2\text{A}^2$$

$$\text{m}^{-3}\text{k}^1\text{s}^0\text{A}^0 \times \text{m}^{-1}\text{k}^{-1}\text{s}^2\text{A}^2 = \text{m}^{-4}\text{k}^0\text{s}^2\text{A}^2$$

$$\sigma^2 = 1.420050989117583 \times 10^{22} \text{ m}^{-4}\text{k}^0\text{s}^2\text{A}^2 \text{ (adds/removes } \mathbf{A}^2 \text{)}$$

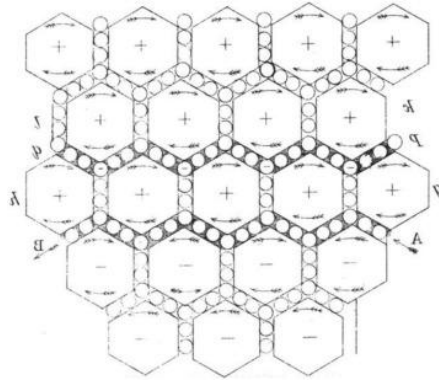
$$\sigma = 1.191658923147720 \times 10^{11} \text{ m}^{-2}\text{k}^0\text{s}^1\text{A}^1 \text{ (adds/removes } \mathbf{A}^1 \text{)}$$

Most calculations need surface charge density squared, but some need surface charge density, so the bottom of this slide shows both for the convenience of anyone who wants to independently verify my calculations. Which one to use depends on whether it is amperes or amperes-squared that you want to add or remove.

A brief history of electromagnetic understanding

elasticity (m/F)

1785/1865 Coulomb/Maxwell



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This slide begins a brief history of the great discoverers who contributed to our current understanding of electromagnetism.

Charles-Augustin de Coulomb first measured the force of attraction and repulsion between electric charges in 1785. Because Coulomb measured the force exerted by pairs of like or unlike charges, discussion of single charges produced seemingly nonsensical measurement units of the square roots of distance and mass. To avoid such nonsensical discussions, the emerging study of electromagnetism spent the next century or so developing entirely new terminology to discuss electromagnetism. That new terminology slowly morphed into modern volts, ohms, amperes, Henries, Farads, etc.

CLICK: During the 1860s, James Clerk Maxwell developed an ingenious mechanical model of the luminiferous aether. His model helped him develop his twenty (or so) equations in 1865, and he realized that Coulomb had discovered how electric charges stretch the luminiferous aether, so he called Coulomb's discovery electric "elasticity". In modern terms, that is meters of stretch per Farad of capacity. Please keep in mind, that is not one electron stretching by a meter to produce a Farad. More likely, Avogadro's number of electrons each stretch by a tiny fraction of their radius.

Remember (from slide 9) that Newton said the elasticity of an isothermal fluid is its pressure.

For this presentation, my synoptic explanations will use modern terminology that often did not exist at the time of the great discoveries. For those who prefer a more chronological and historical deep dive, including the vernacular of the era, I highly recommend the book *The Lightning Tamers: True Stories of the Dreamers and Schemers Who Harnessed Electricity and Transformed Our World* by Kathy Joseph (Kathy Loves Physics of YouTube fame). That book clearly describes the difficulty of discovery, which is much less obvious from the hindsight of my explanations using modern terminology.

A brief history of electromagnetic understanding

elasticity (m/F) 1785/1865 Coulomb/Maxwell

permeability (H/m) 1865/1872 Maxwell/Kelvin

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In 1872, Maxwell's twenty equations led Lord Kelvin to realize Maxwell's magnetism was permeating the aether in the same way that Coulomb's elasticity was stretching it, so he renamed Maxwell's discovery to magnetic "permeability".

Similes: Elasticity is like pressure inflating a balloon. Permeability is like density escaping from a balloon and deflating it. I'm using that analogy, because you should think of the aether as a deformable medium that can be compressed or expanded.

A brief history of electromagnetic understanding

~~elasticity (m/F) — 1785/1865 — Coulomb/Maxwell~~

permeability (H/m) 1865/1872 Maxwell/Kelvin

permittivity (F/m) 1885 Heaviside

impedance (ohms) 1885 Heaviside

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In 1885, Oliver Heaviside thought Lord Kelvin was on the right track, but he took the reciprocal of Kelvin's **elasticity** and called it "**permittivity**" to simplify the math into the modern four Maxwell's equations.

CLICK: Heaviside also added his own new discovery of electromagnetic impedance.

The next slide will demonstrate that these electromagnetic units are the precise physical units that demonstrate with absolute certainty that the luminiferous aether exists.

Coulomb, Maxwell, Kelvin, and Heaviside were still trying to make sense of the incompatible measurement units that made no physical sense to them. It wasn't until SI defined the ampere in 1948 that it first became feasible to convert the incompatible electromagnetic units into the physical units needed to demonstrate that the luminiferous aether was causing electromagnetism and gravity.

Converting between electromagnetic & physical units

electromagnetism	=	physical	EM ← Æ ⇔ Æ ← EM
permittivity (F/m)	=	pressure⁻¹ (m·s²/kg)	$\epsilon_0 = \sigma^2/P \Leftrightarrow P = \sigma^2/\epsilon_0$
elasticity (m/F)	=	pressure (kg/m·s ²)	$k_0 = P/\sigma^2 \Leftrightarrow P = \sigma^2 k_0$
permeability (H/m)	=	density (kg/m ³)	$\mu_0 = \rho/\sigma^2 \Leftrightarrow \rho = \sigma^2 \mu_0$
impedance (ohms)	=	mass flux (kg/m ² s)	$Z_0 = \Phi/\sigma^2 \Leftrightarrow \Phi = \sigma^2 Z_0$
MKSA	=	MKS	MKSA ⇔ MKS

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This slide picks up where ART#142 left off a few weeks ago.

I apologize for moving permittivity up and striking it out, but Lord Kelvin was correct about elasticity and Heaviside was wrong about permittivity. I'm not going to eliminate permittivity, this is just a momentary emphasis on the fact that permittivity may have simplified Heaviside's math but it hindered understanding the role of aether's surface charge density.

The terminology on the left side of this slide identifies the common names of the **MKSA** & **MKS** measurement units. Electromagnetic units are blue, and physical units of the aether are black.

The equations on the right side of this slide show how the squared surface charge density (σ^2) of the aether converts between **MKSA** and **MKS** units.

Elasticity is the more natural way to think about how the pressure of the aether stretches its mass flux outward.

Similarly, permeability is the more natural way to think about how the density of the aether pulls its mass flux inward.

Think about the mass flux as the surface of the balloon, where the surface charge density of the aether resides. The pressure inside the balloon stretches the mass flux outward, and the density outside the balloon compresses the mass flux inward.

In perspective



↑ × dimension	× velocity →		
	mass (kg)	momentum (kg·m/s)	energy (kg·m ² /s ²)
	linear density (kg/m)	mass flow (kg/s)	force (kg·m/s ²)
	area density (kg/m ²)	viscosity (kg/m·s)	surface tension (kg/s ²)
	density (kg/m ³) permeability $\rho = \sigma^2 \mu_0$	mass flux (kg/m ² s) impedance $\Phi = \sigma^2 Z_0$	pressure (kg/m·s ²) elasticity $P = \sigma^2 k_0$

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Permeability is the electromagnetic name for physical density (mass of the aether per volume).

Impedance is the electromagnetic name of physical mass flux (momentum of the aether per volume).

Elasticity is the electromagnetic name for physical pressure (energy of the aether per volume).

Ohm's Law (MSKA)

$V = I \times R$ (voltage = current intensity $\times$ resistance)

Voltage is joules per coulomb, $\text{m}^2\text{k}^1\text{s}^{-3}\text{A}^{-1}$

Current is amperes, $\text{m}^0\text{k}^0\text{s}^0\text{A}^1$

Resistance is volts per ampere, $\text{m}^2\text{k}^1\text{s}^{-3}\text{A}^{-2}$

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This slide adds the **MSKA** units for the most common electromagnetic units of Ohm's Law, as they are specified by NIST and CODATA.

Multiply amperes of current intensity by ohms of resistance and you get volts.

Divide volts by amperes and you get ohms.

Divide volts by ohms and you get amperes.

Now, we're going to convert all of those electromagnetic units into their physical equivalents, so that we can begin exploring how the luminiferous aether produces modern electromagnetic circuits.

Ohm's Law (MKSA-to-MKS)

$$\sigma^2 = 1.420050989117583 \times 10^{22} \text{ m}^{-4} \text{ k}^0 \text{ s}^2 \text{ A}^2$$

$$\sigma = 1.191658923147720 \times 10^{11} \text{ m}^{-2} \text{ k}^0 \text{ s}^1 \text{ A}^1$$

$$\sigma \mathbf{V}: \text{m}^{-2} \text{k}^0 \text{s}^1 \text{A}^1 \times \text{m}^2 \text{k}^1 \text{s}^{-3} \text{A}^{-1} = \text{m}^0 \text{k}^1 \text{s}^{-2} \mathbf{A}^0 \quad (\text{surface tension})$$

$$\mathbf{I}/\sigma: \text{m}^0 \text{k}^0 \text{s}^0 \text{A}^1 \times \text{m}^2 \text{k}^0 \text{s}^{-1} \text{A}^{-1} = \text{m}^2 \text{k}^0 \text{s}^{-1} \mathbf{A}^0 \quad (\text{kinematic viscosity})$$

$$\sigma^2 \mathbf{R}: \text{m}^{-4} \text{k}^0 \text{s}^2 \text{A}^2 \times \text{m}^2 \text{k}^1 \text{s}^{-3} \text{A}^{-2} = \text{m}^{-2} \text{k}^1 \text{s}^{-1} \mathbf{A}^0 \quad (\text{mass flux})$$

$$\mathbf{V} \times \mathbf{I}: \text{m}^2 \text{k}^1 \text{s}^{-3} \text{A}^{-1} \times \text{m}^0 \text{k}^0 \text{s}^0 \text{A}^1 = \text{m}^2 \text{k}^1 \text{s}^{-3} \mathbf{A}^0 \quad (\text{power, watts} = \text{J/s})$$

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Click 1-2: This slide uses the surface charge density of the aether to convert the electromagnetic (MKSA) units to physical (MKS) units. It will do that by dividing or multiplying by σ or σ^2 (as needed) to eliminate A^1 or A^2 from the electromagnetic unit to produce $\mathbf{A}^0$ in the physical unit.

Click 3: This technique multiples σ by **voltage** to cancel the amperes and thereby reveal that electromagnetic voltage is physical surface tension.

Click 4: Similarly, **current intensity** divided by σ cancels the amperes to reveal that electromagnetic current intensity is physical kinematic viscosity.

Click 5: Multiplying σ^2 by **resistance** cancels the amperes to reveal that electromagnetic resistance is physical mass flux.

Click 6: Multiplying **voltage** by **current intensity** doesn't require a conversion factor, because their amperes cancel each other to reveal that electromagnetic watts are physical joules per second. Since voltage was already known to be surface tension, and current intensity was already known to be kinematic viscosity, multiplying those physical attributes produces the same power in joule-seconds.

This isn't shown on the slide, but electromagnetic engineers multiply kilowatts by hours to produce kilowatt-hours of energy consumption. Physicists multiply kilowatt hours by the 3600 seconds in an hour to express each kilowatt-hour of energy consumption as 3.6 megajoules (MJ) of physical energy.

Enhanced perspective



		× velocity →		
↑ × dimension	mass (kg)	momentum (kg·m/s)	energy (kg·m ² /s ²)	
	linear density (kg/m)	mass flow (kg/s)	force (kg·m/s ²)	power (kg·m ² /s ³)
	area density (kg/m ²)	viscosity (kg/m·s)	surface tension (kg/s ²)	kinematic viscosity (m ² /s) current intensity
	density (kg/m ³)	mass flux (kg/m ² s) resistance	pressure (kg/m·s ²)	kinematic viscosity (m ² /s) current intensity

21

Before moving on, I want to remind everyone again that the water in Newton's bucket simultaneously has all of the attributes depicted by this table.

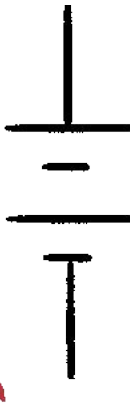
Table 2-1 from my book, *The Luminiferous Aether: primary substance of the universe*.

I've added the electromagnetic units of **voltage**, **current intensity**, and **resistance** to this table. Notice that because current intensity has no mass of its own, it is a diagonal transition of one dimension vertically by one velocity horizontally to produce its measurement units of m²/s (**kinematic viscosity**).

Starting at the bottom center, resistance times current intensity is voltage, and voltage times current intensity is power.

I didn't attempt to draw this on the already cluttered table, but you can imagine a Z-axis going through the page to multiply or divide by the necessary power of surface charge density to transform between the physical and electromagnetic perspectives of the same attributes.

Simple circuitry (voltage source)

$$\checkmark \left(\frac{\text{kg}}{\text{s}^2} \right) \text{ surface tension}$$


$$\text{Eq 4-3: } \gamma = PR_L = \rho \varkappa$$

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The black symbol just to the right of center is the schematic diagram symbol for a voltage source.

It is meant to depict the plates of a voltaic pile, which was the first chemical battery (invented in 1800 by Alessandro Volta).

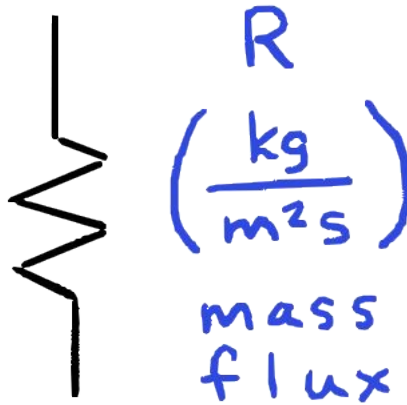
(According to Gemini AI,) voltaic piles were constructed by stacking alternating pairs of dissimilar metal discs (such as zinc and copper, or zinc and silver) separated by brine- or acid-soaked cardboard, cloth, or leather.

Each additional plate added to the pile boosts the surface tension (voltage) of the circuit a little higher.

Eq 4-3 (from my book, *The Luminiferous Aether*,) reveals that the surface tension of the aether is the pressure of the aether multiplied by the luminiferous radius of the substance involved, or the density of the aether multiplied by the gravitational parameter of the substance involved.

This is the master equation that ties electro-magno-gravity together. Gamma (voltage/surface-tension) = aether pressure (volts/surface-tension per distance) times luminiferous radius = aether density times gravitational parameter (acceleration of gravity times distance-squared).

Simple circuitry (resistor)



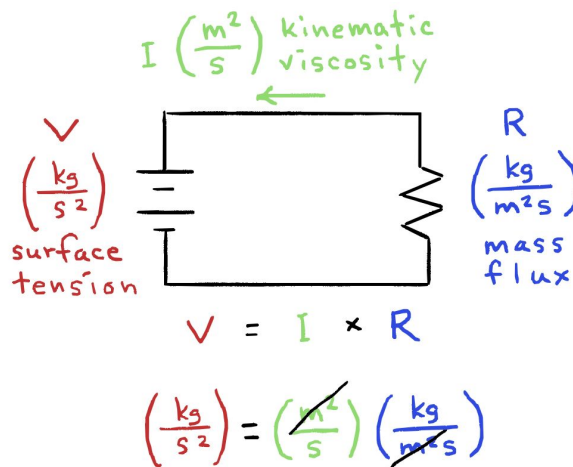
23

The black symbol just to the left of center is the schematic diagram symbol for a resistor.

The jagged lines are meant to depict an impeded or restricted current flow. Alternatively, it can be viewed as a staircase that the surface tension gradient must climb or descend as it crosses the resistor. Mass flux represents the undisturbed state of the luminiferous aether. It is the momentum of the aether per volume, so mass flux impedes current flow by being in its way.

Resistance causes collisions among aether substance (neutrinos, or whatever) that are dissipated by the resistor as heat.

Ohm's Law for a simple circuit



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Here is the schematic diagram of a simple electrical circuit using the components we just discussed on the prior slides. Notice that current intensity is a massless transition (distance times velocity, or area per second) between attributes, just like it was for the table on slide 21.

You can easily see that the surface tension (voltage) does indeed equal the kinematic viscosity (current intensity) times the mass flux (resistance).

Considering that mass flux is the chaotic state of random aether, you can now realize why kinematic viscosity causes electric current to “flow”. The area of the kinematic viscosity cancels the area of the mass flux. Those cancellations are shown below the schematic diagram to make it easy to verify the math.

The kilograms per second (remaining after cancelling the area) is mass flow between the voltage source and the resistor. Similarly, the per second of the kinematic viscosity (that remains after cancelling the area) combines with the mass flow to produce mass flow per second, which is the surface tension of the voltage source. Consequently, the momentum per volume that is the mass flux of the resistance pushes a mass flow through the kinematic viscosity, toward the surface tension of the voltage source.

That's a lot to take in! Are there any questions about this, before we add some more electronic components to our vocabulary so that we can examine some more complex circuit diagrams?

Capacitance & inductance

Farads: $\text{m}^2\text{k}^{-1}\text{s}^2\text{A}^0$ (area per surface tension = area per volt)

Henries: $\text{m}^{-2}\text{k}^1\text{s}^0\text{A}^0$ (area density = mass per area)

Click 1: From slide 16, the reciprocal of pressure is Farads per meter.

Thus, Farads of capacitance is area per surface tension (which is area per volt).

Click 2: From slide 16, density is Henries per meter.

So, Henries of inductance is area density (which is mass per area).

Simple circuitry (capacitor)



The schematic symbol for a capacitor is a black symbol consisting of two parallel horizontal lines of equal length, connected by two vertical lines of equal length. The symbol is positioned to the left of the handwritten equations.

$$C = \frac{\text{area}}{\text{surface tension}}$$

Handwritten in green, the equation shows the unit for capacitance, C , as $\left(\frac{\text{m}^2 \text{s}^2}{\text{kg}}\right)$. Below this, the equation is written as $\frac{\text{area}}{\text{volt}} = \frac{\text{area}}{\text{surface tension}}$.

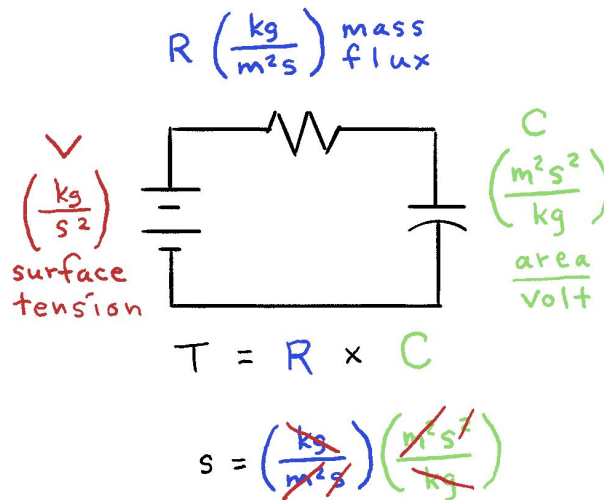
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The black symbol just to the left of center is the schematic diagram symbol for a capacitor.

That symbol is meant to depict a break in the circuit consisting of the plates of a capacitor with an insulator between them. The insulator is called a dielectric. Depending on the working voltage needed for the capacitor, common dielectrics are air, glass, oil, paper, and plastic.

A capacitor opposes changes to surface tension (kg/s^2 , voltage) by spreading high charge and discharge current intensity (m^2/s , kinematic viscosity) across the area of its plates.

A simple RC (resistor-capacitor) circuit



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Here is another simple electrical circuit. This one adds the capacitor from slide 26 to the circuit from slide 24.

Current flows in this circuit only while the capacitor is charging or discharging to the level of the surface tension provided by the battery. The mass flux of the resistor restricts how quickly current can flow into the capacitor. Just as for the simple resistor-capacitor (RC) circuit of slide 24, the kinematic viscosity (current intensity) limits how quickly mass can flow through the resistor.

Mass flux (resistance) times area per surface tension (capacitance) determines how many seconds it will take to charge the capacitor to the surface tension (voltage) of the battery. Current stops flowing when the surface tension of the capacitor equals the surface tension of the battery.

Simple circuitry (inductor)

$$L \left(\frac{\text{kg}}{\text{m}^2} \right) \frac{\text{mass}}{\text{area}}$$



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The black symbol at the bottom of this slide is the schematic diagram symbol for an inductor.

That symbol is meant to depict a magnetic field developing within a coil of wire. This symbol is intentionally similar to a resistor, because it restricts **changing** current.

It's more like a water slide than the staircase of the inductor. A surface tension (kg/s^2 , voltage) difference can build across it initially, but that surface tension will rapidly decrease to no difference as the kinematic viscosity (m^2/s , current intensity) stabilizes.

A simple LC (inductor-capacitor) circuit

$$L \left(\frac{\text{kg}}{\text{m}^2} \right) \frac{\text{mass}}{\text{area}}$$

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$C \left(\frac{\text{m}^2 \text{s}^2}{\text{kg}} \right) \frac{\text{area}}{\text{volt}}$$

$$T^2 = L \times C$$

$$\text{s}^2 = \left(\frac{\cancel{\text{kg}}}{\cancel{\text{m}^2}} \right) \left(\frac{\cancel{\text{m}^2} \text{s}^2}{\cancel{\text{kg}}} \right)$$

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This new circuit replaces the resistor of the prior circuit with an inductor. It also replaces the battery with an alternating current (AC) source.

The schematic symbol for the AC generator is meant to depict a constantly changing current.

In conjunction with a capacitor, an inductor forms an oscillating circuit with a resonant frequency. That happens because the inductor opposes current flow into the capacitor as it charges, but then effectively becomes a short circuit when the capacitor is fully charged. The capacitor then dumps its charge back into the inductor, but the inductor delays the current flow back to the generator. Essentially, the capacitor and inductor keep dumping delayed current back and forth between each other, and that's what causes the circuit to oscillate at the resonant frequency.

Capacitive reactance, surface tension (V) lags

$$\text{m}^2\text{k}^{-1}\text{s}^2 \times \text{m}^{-2}\text{k}^1\text{s}^{-1} = \text{m}^0\text{k}^0\text{s}^1$$

$$C \times \text{mass flux (X)} = \text{lag seconds of } \mathbf{D} = \epsilon_0 \mathbf{E}$$

30

Capacitors cause surface tension (voltage) changes to lag, without hindering kinematic viscosity (current intensity).

Capacitance multiplied by mass flux is seconds of phase lag of surface tension (voltage).

IOW, it differs from impedance only by seconds (lag of the **D** & **E** fields). It is mass flux (impedance) of the aether with reduced surface tension (voltage) lagging slightly behind kinematic viscosity (current intensity), thereby causing a slight tilt of the aether surface toward higher pressure.

Inductive reactance, kinematic viscosity (I) lags

$$m^{-2}k^1s^0 \times m^2k^{-1}s^1 = m^0k^0s^1$$

$$L / \text{mass flux (X)} = \text{lag seconds of } \mathbf{B} = \mu_0 \mathbf{H}$$

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Inductors cause kinematic viscosity (current intensity) changes to lag, without hindering surface tension (voltage).

Inductance divided by mass flux is seconds of phase lag of kinematic viscosity (current intensity).

IOW, it differs from impedance only by seconds (lag of the **B** & **H** fields). It is mass flux (impedance) of the aether with reduced kinematic viscosity (current intensity) lagging slightly behind surface tension (voltage), thereby causing a slight tilt of the aether surface toward higher density.

Physical **mass flux** = electromagnetic ohms

Resistance:	$R = V / I$	$\text{m}^{-2}\text{k}^1\text{s}^{-1}$
Capacitive reactance:	$X_C = (2\pi fC)^{-1}$	$\text{m}^{-2}\text{k}^1\text{s}^{-1}/2\pi$
Inductive reactance:	$X_L = 2\pi fL$	$2\pi \cdot \text{m}^{-2}\text{k}^1\text{s}^{-1}$
Impedance:	$Z^2 = R^2 + X_C X_L = k_0 \mu_0 = P\rho$	$(2^{-1/2}\text{m}^{-2}\text{k}^1\text{s}^{-1})^2$

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Mass flux of the luminiferous aether reacts to how it is being perturbed.

Click 1: Resistance is mostly just pure random mass flux. That qualifier “mostly” is there because every resistor has a small amount of capacitive and/or inductive reactance at certain frequencies.

Click 2: Capacitors oppose surface tension (voltage) changes. Thus, capacitive reactance (slightly less random mass flux) is inversely proportional to frequency. As the frequency of perturbations increases, opposition to kinematic viscosity (current intensity) decreases.

Click 3: Inductors oppose kinematic viscosity (current intensity) changes. Thus, inductive reactance (slightly less random mass flux) is directly proportional to frequency. As the frequency of perturbations increases, opposition to kinematic viscosity (current intensity) increases.

Click 4: Impedance represents the combined hindrance to mass flow by all possible perturbations of the aether. It is stable because it represents the ability of the aether’s mass flux to mediate all density and pressure perturbations.

FractalWoman has a nice animation [An Aether Model of Electricity : The Missing Secret of Magnetism](https://www.youtube.com/watch?v=T-tEAL1eU_0), where she discusses surface wave phenomena. For those of you who watch her video, please realize that superfluid pressure (residing above the mass flux) behaves like a different medium from superfluid density (residing below the mass flux). That will assist with understanding why surface charge density matters.

Are there any questions?

Q & A

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Are there any questions?